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| | **Techncial Specification for Sliding roof covers - Amaala project at Intake pump house** | NIHVA remarks | |
| | Sliding roof covers, referred to as retractable or sliding roof systems, are structures that can open or close to cover outdoor or indoor spaces, providing flexibility for light, ventilation, and weather protection. | | |
| | Note: Provided snap below & details shared are indicative. However, vendor can share their proposal for our review | | |
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| Sr | **Technical Specification: Polycarbonate Opaque Sliding Roof Cover with Aluminum Frame** | | |
| | **General Description** | | |
| | **Roof Cover Type**: Sliding roof cover consisting of two panels | Drawing attached | Remarks |
| | **Covering**: | | |
| | **Type**: Polycarbonate opaque sheets | OK | colour |
| | **Thickness**: 6mm to 10mm (to be finalized based on structural analysis) | 6mm | |
| | **Properties**: High impact resistance, lightweight, UV-stabilized, and opaque to block sunlight | OK | |
| | **Frame**: | | |
| | **Type**: Aluminum alloy (6063-T6 or equivalent) | OK | |
| | **Properties**: Corrosion-resistant, strong, and suitable for high-temperature environments | | |
| | **Dimensions** | | |
| | **Roof Cutout Size**: Refer shared drawing for actual dimensions | Drawing attached | |
| | **Panel Configuration**: shall be proposed by vendor | AS per drawing | |
| | **Panel Operation** | | |
| | **Mechanism**: Manually operated | manually Operated | |
| | **Structural Support** | | |
| | **Frame Structure**: Aluminum frame with reinforced joints to withstand wind loads and thermal expansion | OK | |
| | **Mounting**: Securely anchored to the RCC slab using stainless steel bolts and weatherproof seals to prevent water ingress | OK | |

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| | **Sealing and Weatherproofing** | | |
| | **Seals**: EPDM rubber or silicone seals between panels and frame to prevent dust and water entry | Panel will be in NIHVA scope | Frame and panel difference ? |
| | **Drainage**: Integrated gutter system with downspouts to manage rainwater runoff | OK | |
| | **Environmental Considerations** | | |
| | **Temperature Resistance**: Materials rated for temperatures ranging from -20°C to 50°C, accommodating Saudi Arabia's climate | OK | |
| | **UV Resistance**: UV-stabilized polycarbonate sheets to prevent degradation from prolonged sun exposure | OK | |
| | **Dust Resistance**: Smooth surfaces and sealed edges to minimize dust accumulation and facilitate cleaning | OK | |
| | **Safety Features** | | |
| | **Locking Mechanism**: Automatic locking system when panels are closed to ensure security | Ok | Need electrical power and control panel hence we will suggest you to lock mnaually |
| | **Emergency Stop**: Manual override and emergency stop button for the motorized system | ???? | manual driven from top |
| | **Maintenance** | | |
| | **Access**: Removable panels or access hatches for periodic cleaning and maintenance | ??? | |
| | **Durability**: Corrosion-resistant materials selected for long-term performance with minimal upkeep | OK | |

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| | **Compliance** | | |
| | **Standards**: Designed to comply with the Saudi Building Code (SBC) and relevant ASTM standards | Our Design requirements comply with combination of relevant Saudi Building Codes (SBC) (which govern structural, architectural and loads requirements) and material standards for roofing and components | e.g. - SBC 201 (General), SBC 301 (Loads), SBC 306/304 (Structural) |
| | **Certifications**: Manufacturer to provide material and system certifications confirming quality and suitability | OK | |
| | **Loading** | | |
| | **150kg/m2** shall be considered over the panels as Live load | OK | |